

Today's Topics

Revision of Servlets

Continue with Servlet JDBC Integration

Page Navigation

Server side state Management using Cookie n later HttpSession

Revision

What is Java EE/Jakarta EE ?

Specs for primary services required to build any enterprise app.

eg : Servlet API . JSP ,Connection pooling , security,JPA, web services...

Why J2EE 1.8/Java EE 8 /Jakarta EE 8

1. server independence
2. supports variety of clnts
3. J2EE server supply rdy made imple of prim services

What makes Java EE , server independent ?
separation between specs n imple.

Who provides Specs of prim services in J2EE / Java EE : Oracle/Sun / Eclipse Foundation

Implementation : Java EE server vendors

eg: For Servlets ,

Servlet specs -- servlet-api.jar (server independent)

Imple. is left to vendors --Apache Tomcat -- catalina.jar

	Specification	Implementation
Servlet	servlet-api-jar	catlina.jar
JSP	jsp-api-jar	jasper.jar

For JSP -

JSP specs -- jsp-api.jar (server independent)

--coming from Java EE

Imple. is left to vendors --Apache Tomcat -- jasper.jar

Popular vendors & their products

(Apache : Tomcat / Tomee, Oracle --Glassfish , Jboss , weblogic....)

1. What is WC ?

Server side JVM , provides a run time environment to web apps

Jobs ---

manages life cycles of web comps , creates server req n resp objs,manages session tracking , manages concurrency, navigation life cycle of dynamic web components ...

2. What is web.xml? --- deployment descriptor of a web app(per web app)

populated by --prog (IDE)

who reads it -- WC

when ---exactly once @ web app dep time

what does it consist of --- xml tags (=> deployment instructions)

eg : welcome file , servlet deployment tags, sec tags, session tags...

Mandatory till Java EE 1.5 , optional later.

J2EE compliant folder structure of dynamic web application

1. WEB-INF : private

web.xml

<classes> : .class (hidden) --compiled java classes

<lib> : .jar

eg : JDBC driver , Spring , hibernate JARs

2. root folder : public -- src/main/webapp (/) -- html , js, css , jsp

3. src/main/java : java src files

3. Why servlets : Adds dyn nature to the web app

Job list

1. req processing

2.dyn resp gen.

3. B.L(Business logic)

4. Managing DAO layer

5. Page navigation

What is a servlet ?

Java class , no main method , dyn web component whose life cycle is managed by WC

life cycle methods : init , service , destroy.

1. Revise Servlet API details --

Servlet API (IMPORTANT)

1. javax.servlet.Servlet i/f

5 methods : 3 important life cycle methods

public void init(ServletConfig config) throws ServletException

public void destroy()

public void service(ServletRequest rq,ServletResponse rs) throws ServletException, IOException

2. Imple. class : javax.servlet.GenericServlet : abstract class

concret methods --init , destroy

abstract --service

3. For HTTP specific :

extended by : javax.servlet.http.HttpServlet : abstract class

no abstract method !

Has implemented : public void service(ServletRequest rq,ServletResponse rs) throws ServletException, IOException

---> protected void service(HttpServletRequest rq,HttpServletResponse rs) throws ServletException, IOException

---> dispatcher -->

chks HTTP method

get --> protected void doGet(HttpServletRequest rq,HttpServletResponse rs) throws ServletException,

IOException
post --> doPost
put --> doPut
delete --> doDelete

Deployment of the servlet

1. Via annotation

eg : @WebServlet(value="/hello")

public class HelloWorldServlet extends HttpServlet {...}

Meaning : A class level annotation , for WC(run time env) , to add the request mapping between incoming request (url-pattern) n the servlet.

Map : will be created by WC , at web app deployment time

key -- /hello

value -- Fully qualified servlet cls name (eg : pages.HelloWorldServlet)

URL : http://host:port/day2.1/hello

URI : /day2.1/hello

URL pattern : /hello

OR

2. Using XML tags

How to deploy a servlet w/o annotations --- via XML tags

web.xml

```
<servlet>
  <servlet-name>abc</servlet-name>
  <servlet-class>pages.SecondServlet</servlet-class>
</servlet>
<servlet-mapping>
  <servlet-name>abc</servlet-name>
  <url-pattern>/test2</url-pattern>
</servlet-mapping>
```

WC : map

key : /test2

value : pages.SecondServlet

eg URL --http://host:port/day2.1/hello

At the time of web app deployment ---WC tries to populate map of request mappings , from XML tags (from web.xml). Later ---it will check for @WebServlet annotation

1. What is default loading policy of WC for servlets (when will WC start life cycle of the servlet)

Ans - after any clnt sends the 1st request , indicating lazy loading .

Can you change it to eager ? YES

HOW ?

1. Using the attribute in the @WebServlet

- loadOnStartup

default value=-1 => lazy loading

If you supply > 0 =>eager loading

You can also specify the order of loading.

Use case : Only in case of time consuming init methods

eg : creating DAO instance from init().

OR

2. Using xml tags

Add Within <servlet> tag

<servlet-name> ...</>

<servlet-class> ..</>

<load-on-startup>1</...>

2. Servlet API to read request params sent from the client in the servlet ?

javax.servlet.ServletRequest i/f methods

1. public String getParameter(String paramName)

2. public String[] getParameterValues(String paramName)

Case study (online voting)

Objective : Servlet --JDBC Integration

Steps

0. Copy JDBC Connector JAR(mysql connector) : <WEB-INF>/lib OR <tomcat_home> / lib
Already done in pre-configured Tomcat.

Layers involved : Servlet --DAO (Uses DBUtils - open n close cn) --POJO/Entities -- DB

Refer case study : Online Voting

Design (refer : servlet-jdbc-integration.png)

1. Table : users

2. POJO : User

3. Create DBUtils :

3.1 open fixed DB connection : opens the fixed con to DB n return it to the caller.

3.2 close connection

3.3 Get connection

4. DAO : UserDao i/f n implementation class

6. welcome page : login.html(post) --> LoginServlet

7. --> LoginServlet

7.1 --init : dao instance , destroy : dao's cleanup , doPost : auth user method

8. Centralized exc handling in Servlets

How ?

@Override

```
public void init() throws ServletException
```

```
{
```

```
try {
```

```
    open connection
```

```
    create dao inst
```

```
} catch(Exception e)
```

```
{
```

```
//re throw the exception back to WC , so that WC doesn't continue to service.
```

```
    throw new ServletException("err in init" +getClass(),e);
```

```
}
```

```
}
```

API of javax.servlet.ServletException

Constructor :

```
public ServletException(String message,Throwable rootCause)
```

Objective :

Complete login--logout flow in current app

(login.html -- LoginServlet --successful login --check the role

If admin --redirect to admin servlet

If voter , who has not yet voted -- redirect to candidate list page

If voter , who has already voted -- redirect to logout page

9. How to redirect ?

Refer : "day2-data\day2_help\page navigation\page navigation techniques.txt"

10. Do you see any problems after redirect ?

After redirecting the clnt to the next page in the NEXT request ,

while accessing req params from next page (request.getParameter) : got the null value .

Problem cause : HTTP is stateless inherently (i.e can't remember clnt's conversational state across multiple requests)

Solution : State management on Server side , using Session Tracking